

ZAYAAN BHANWADIA

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EDUCATION

University of Toronto

Honours B.Sc. in Computer Science (Co-op) | GPA: 3.7/4.0

Sept. 2025 – Dec. 2028

Toronto, ON

EXPERIENCE

Software Developer Intern (*Incoming Full-Stack*)

FreshBooks

Sept. 2026 – Dec. 2026

Toronto, ON

- Incoming co-op developing Python backend microservices and React UI for a platform serving **10M+** users.

Software Engineer Intern (*Backend AI*)

FlyRank AI

July 2026 – Present

Toronto, ON

- Built a centralized embeddable widget platform routing form submissions to a dashboard via a one-line script tag.
- Secured public endpoint with validation, rate limiting, spam filtering, and a geolocation fallback, passing **80+** tests.
- Prevented duplicate charges under **50 concurrent requests** by deduplicating billable events at the database level.
- Synced Stripe payment updates through secure, idempotent webhooks, eliminating duplicate records and manual fixes.

Software Engineer Intern (*Full-Stack*)

KorraNet Creative (Riipen Program)

May 2026 – July 2026

Remote

- Optimized Meta OAuth token handling and storage, reducing token refresh failures by **40%+** for AI integrations.
- Eliminated all hardcoded secrets across the codebase by restructuring ingestion into secure environment management.
- Reduced system crashes and failed API calls by **85%+** through adding API error handling and timeout prevention.
- Implemented logging across onboarding and authentication flows to isolate bugs and reduce debugging time.

PROJECTS

Embedded Key-Value Store | C++, POSIX Threads, Linux

- Built an embedded C++ database with data sharding to prevent global lock bottlenecks and improve throughput.
- Implemented shared locks for non-blocking reads across **4 threads**, preventing race conditions during writes.
- Designed data persistence by writing updates to a log file before memory to ensure crash recovery and durability.
- Created a file compaction routine to safely clean up keys, **reducing disk usage by 45%+** without pausing the app.

Context Sync Extension | TypeScript, VS Code Extension API

- Developed a VS Code extension with **500+** downloads that automates AI chat context sharing across environments.
- Designed a graph-based Markdown schema to maximize context density per token across AI assistant sessions.
- Integrated GitHub Copilot, OneDrive, and Google Drive to automate syncing and enable one-click environment setup.

Clinical AI Assistant (Top 10 Projects @ GenAI Genesis) | Python, SQL, OpenCV

- Automated patient check-ins and intake with an AI-powered clinical assistant, structuring records into a SQL schema.
- Engineered a live monitoring dashboard with computer vision, achieving **98%+** accuracy for real-time fall detection.
- Developed a spatial visualization algorithm that generates 3D labelled hospital models by extruding floor plans.

Table Tennis Referee (Finalist @ Hack Canada) | Python, YOLO

- Achieved **90%+** ball tracking accuracy by engineering and training a live computer vision system using OpenCV.
- Automated real-time score tracking by building game-state logic on coordinates from YOLOv8 object detection.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, SQL, Bash, HTML/CSS

Libraries/Frameworks: FastAPI, Node.js, Flask, SQLAlchemy, React.js, OpenCV

Developer Tools: Linux, Docker, Git, GitHub, PGAdmin, VS Code, Jira/Confluence

Cloud/Databases: PostgreSQL, IBM Cloud, Google Cloud Platform, Vercel, Vultr, Railway, Supabase